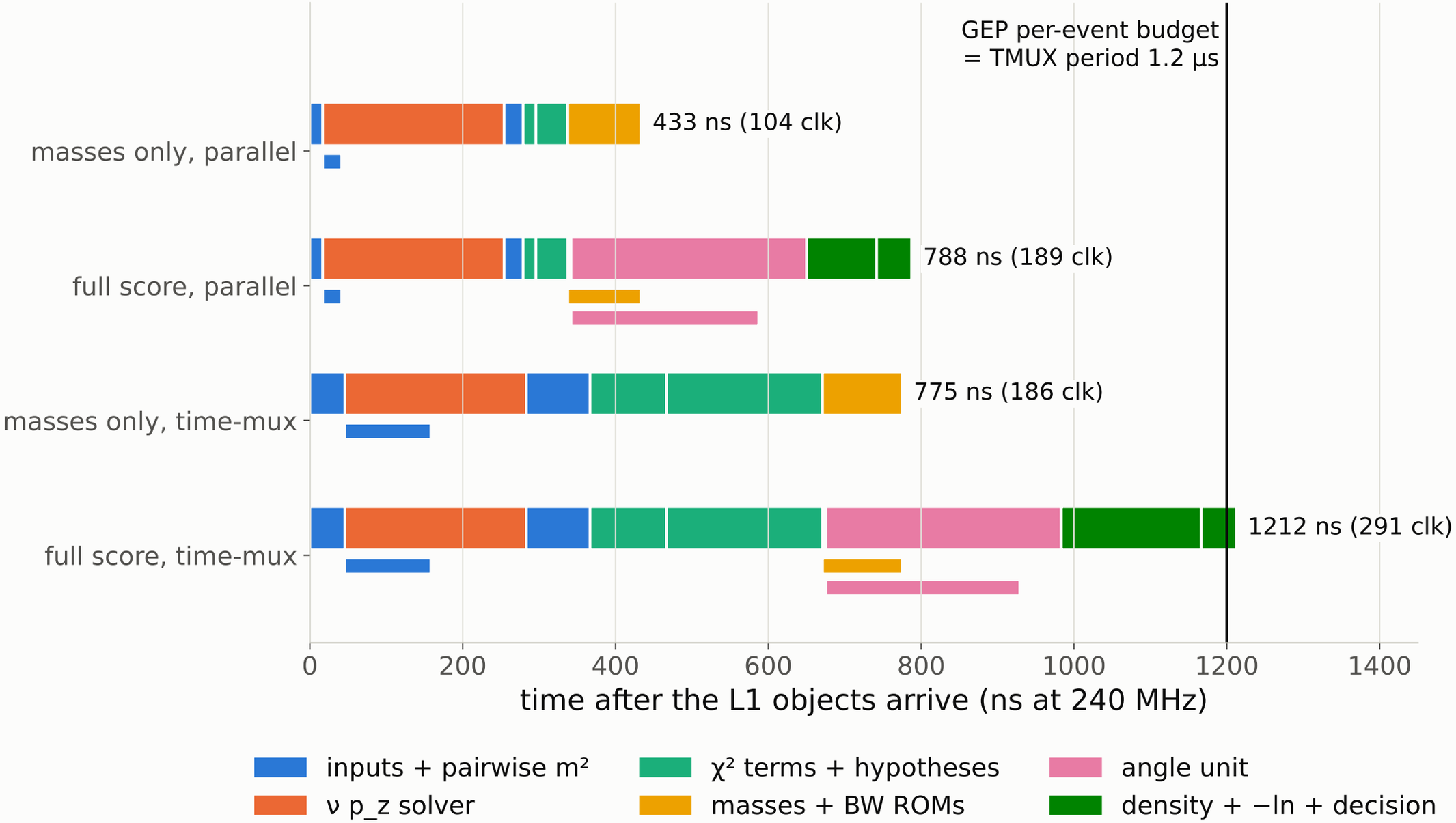

Thick bars: critical path; thin bars: blocks running concurrently. The whole ATLAS L0 budget of 10 µs is off scale (8–23× the totals). Analytic pipeline depths (DSP58 3 clk, dot-product cascades, recip/rsqrt/sqrt units 12–18 clk, CORDIC W+1 clk); N = 6, W = 16, K = 256. At 360 MHz all four designs scale by 2/3 (full score, time-mux: 808 ns).